2.2 Integration Worksheet

1. Find $\int \sin^2(x)\cos^2(x)\ dx$.
   Since the powers of sine and cosine are both even we will use the double angle identities. We have:
   $$\int \sin^2(x)\cos^2(x)\ dx = \int \frac{1}{2}\left(1-\cos(2x)\right)\frac{1}{2}\left(1+\cos(2x)\right)dx = \frac{1}{4}\int\left(1-\cos^2(2x)\right)dx$$
   $$=\frac{1}{4}\int\left(1-\frac{1}{2}\left(1+\cos(4x)\right)\right)dx = \frac{1}{4}\int\left(\frac{1}{2}-\frac{1}{2}\cos(4x)\right)dx = \frac{1}{8}\int\left(1-\cos(4x)\right)dx = \frac{1}{8}\left(x-\frac{1}{4}\sin(4x)\right)+C.$$

2. Find $\int \cos^{1/3}(x)\sin(x)\,dx$.

   Let $u=\cos(x)$, then $du=-\sin(x)\,dx \rightarrow -du=\sin(x)\,dx$.
   Therefore, $\int \cos^{1/3}(x)\sin(x)\,dx = \int -u^3du = -\frac{1}{4}u^4+C = -\frac{1}{4}\cos^4(x)+C$

3. Find $\int \sin^3(x)\cos^2(x)\,dx$.

   Let $u=\cos(x)$, then $du=-\sin(x)\,dx \rightarrow -du=\sin(x)\,dx$.
   Therefore,
   $$\int \sin^3(x)\cos^2(x)\,dx = \int -\sin^2(x)u^2du = \int -\left[1-\cos^2(x)\right]u^2du = \int -\left[1-u^2\right]u^2du = \int\left[u^4-u^2\right]du$$
   $$=\frac{1}{3}u^3+\frac{1}{5}u^5+C = \frac{1}{3}\cos^3(x)+\frac{1}{5}\cos^5(x)+C$$

4. Find $\int_0^{\pi} \sin^3(x)\cos^3(x)\,dx$.

Let $u=\cos(x)$, then $du=-\sin(x)\,dx \rightarrow -du=\sin(x)\,dx$.

The bounds become $u(0)=1$ & $u(\pi)=-1$

Therefore,

$$\int_0^{\pi} \sin^3(x)\cos^3(x)\,dx=\int_1^{-1} -\sin^2(x)u^3du=\int_1^{-1} -\left[1-\cos^2(x)\right]u^3du=\int_{-1}^{1}\left[1-u^2\right]u^3du=\int_{-1}^{1}\left[u^3-u^5\right]du$$

$$=\frac{1}{4}u^4-\frac{1}{6}u^6\Big|_{-1}^{1}=\left(\frac{1}{4}-\frac{1}{6}\right)-\left(\frac{1}{4}-\frac{1}{6}\right)=0$$

OR

Let $u=\sin(x)$, then $du=\cos(x)\,dx$.

The bounds become $u(0)=0$ & $u(\pi)=0$

Therefore,

$$\int_0^{\pi} \sin^3(x)\cos^3(x)\,dx=\int_0^{0} u^3\cos^2(x)\,du=\int_0^{0} u^3\left[1-\sin^2(x)\right]du=\int_0^{0}\left[1-u^2\right]u^3du=0$$

5. Find $\int \tan^3(x)\sec^5(x)\,dx$.

$$\int \tan^3(x)\sec^5(x)\,dx=\int \tan^2(x)\sec^4(x)\sec(x)\tan(x)\,dx=\int\left[\sec^2(x)-1\right]\sec^4(x)\sec(x)\tan(x)\,dx$$

We will now use a u-sub. Let $u=\sec(x)$, then $du=\sec(x)\tan(x)\,dx$. Then we have:

$$\int\left[u^2-1\right]u^4du=\int\left[u^6-u^4\right]du=\frac{1}{7}u^7-\frac{1}{5}u^5+C=\frac{1}{7}\sec^7(x)-\frac{1}{5}\sec^5(x)+C$$

6. Find $\int_0^{\pi/4} \tan(x)\sec^3(x)\,dx$.

Let $u=\sec(x)$, then $du=\sec(x)\tan(x)\,dx$

Our bounds become $u(0)=\sec(0)=1$ and $u\left(\frac{\pi}{4}\right)=\sqrt{2}$

Therefore,

$$\int_0^{\pi/4} \tan(x)\sec^3(x)\,dx=\int_1^{\sqrt{2}} u^2du=\frac{1}{3}u^3\Big|_1^{\sqrt{2}}=\frac{1}{3}2^{3/2}-\frac{1}{3}=\frac{1}{3}\left[2^{3/2}-1\right]$$

7. Find $\int \tan^3(x)\sec^2(x)\,dx$

Let $u=\tan(x)$, then $du=\sec^2(x)\,dx$
Therefore,

$$\int \tan^3(x)\sec^2(x)\,dx=\int u^3du=\frac{1}{4}u^4+C=\frac{1}{4}\tan^4(x)+C$$

8. Find $\int \cot^4(x)\csc^4(x)\,dx$

Let $u=\cot(x)$, then $du=-\csc^2(x)\,dx \;\rightarrow\; -du=\csc^2(x)\,dx$
Therefore,

$$\int \cot^4(x)\csc^4(x)\,dx=\int -u^4\csc^2(x)\,du=\int -u^4\left[1+\cot^2(x)\right]du=\int -u^4\left[1+u^2\right]du=\int\left[u^4-u^6\right]du$$
$$=\frac{1}{5}u^5-\frac{1}{7}u^7+C$$
$$=\frac{1}{5}\cot^5(x)-\frac{1}{7}\cot^7(x)+C$$

9. $\int \sec^4(x)\,dx$

Let $u=\tan(x)$ and then $du=\sec^2(x)\,dx$.
Therefore,

$$\int \sec^4(x)\,dx=\int \sec^2(x)\,du=\int\left[\tan^2(x)+1\right]du=\int\left[u^2+1\right]du=\frac{1}{3}u^3+u+C$$
$$=\frac{1}{3}\tan^3(x)+\tan(x)+C$$

10. $\int \tan^4(x)\,dx$

$$\int \tan^4(x)\,dx = \int \tan^2(x)\tan^2(x)\,dx = \int\left[\sec^2(x)-1\right]\left[\sec^2(x)-1\right]dx = \int\left[\sec^4(x)-2\sec^2(x)-1\right]dx$$
$$= \int \sec^4(x)\,dx - \int\left[2\sec^2(x)+1\right]dx$$

On the first integral we let $u=\tan(x)$ and then $du=\sec^2(x)\,dx$.
Therefore,

$$\int \sec^4(x)\,dx - \int\left[2\sec^2(x)+1\right]dx = \int \sec^2(x)\,du - \left[2\tan(x)+x\right]+C = \int\left[\tan^2(x)+1\right]du - \left[2\tan(x)+x\right]+C$$
$$= \int\left[u^2+1\right]du - \left[2\tan(x)+x\right]+C = \frac{1}{3}u^3+u-\left[2\tan(x)+x\right]+C$$
$$= \frac{1}{3}\tan^3(x)+\tan(x)-\left[2\tan(x)+x\right]+C$$
$$= \frac{1}{3}\tan^3(x)-\tan(x)-x+C$$